

Introduction to Systems Thinking



**“The significant problems we face today
cannot be solved at the same **level of
thinking** at which they were created.”**

Albert Einstein

WHAT IS SYSTEMS THINKING?

Founded in 1956 by MIT professor, Jay Forrester.

System thinking allows people to make their **understanding of social system** explicit and improve them in the **same way that people use engineering principles** to improve their understanding of mechanical system.

It is use to:-

- Examining how we create our own problems
- Seeing the big picture
- Structure influences performance

Basic Properties of a System

system must consist of :

- ✓ Elements or parts
- ✓ Interconnectedness & Interactions
- ✓ Function or purpose

□ **Examples:** a business, football team;
digestive system; school; faculty, city;
corporation; animal; tree; etc.

SYSTEM THINKING APPROACH

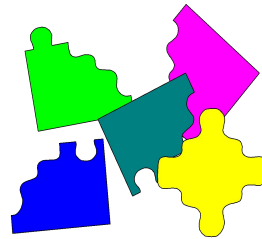
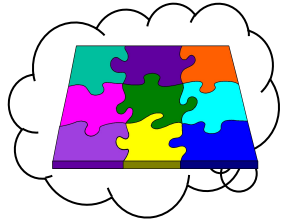
Traditional analysis	Systems thinking
Traditional analysis focuses on the separating the individual pieces of what is being studied; in fact, the word “analysis” actually comes from the root meaning “to break into constituent parts.	Systems thinking, in contrast, focuses on how the thing being studied interacts with the other constituents of the system—a set of elements that interact to produce behavior—of which it is a part.

Examples of area System Thinking has proven its value include:

- Complex problems that involve helping many actors see the “big picture” and not just their part of it
- Recurring problems or those that have been made worse by past attempts to fix them
- Issues where an action affects (or is affected by) the environment surrounding the issue, either the natural environment or the competitive environment
- Problems whose solutions are not obvious

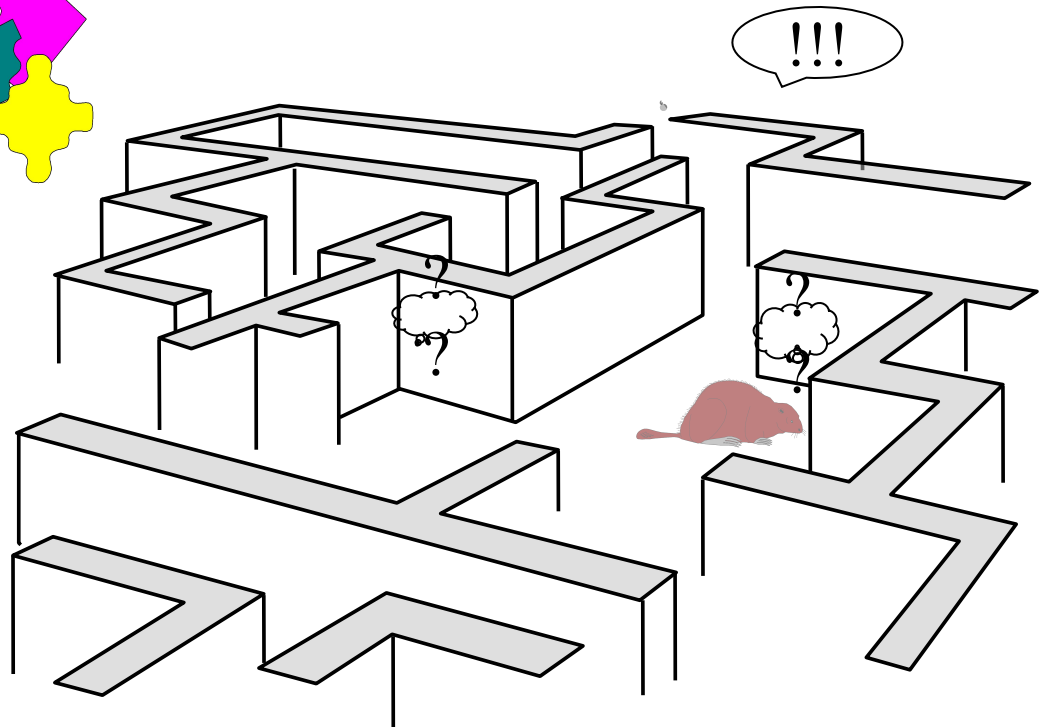
Systems Thinking

Learning to see the world systemically
Encourages us to see the whole as well as the parts.



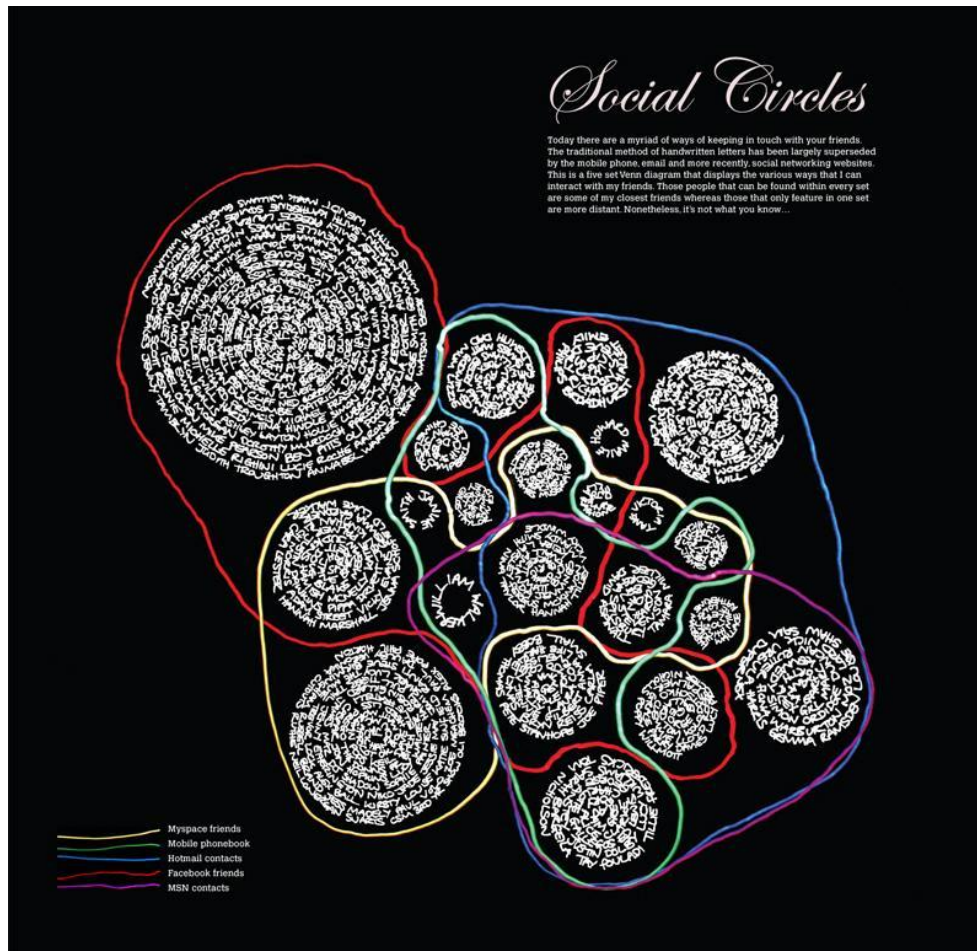
WHOLE $\neq$ Σ PARTS
 $\Rightarrow$ Holistic Thinking

**Multiple (often)
restricted views**



Systems Thinking...

Helps us explore interdependencies and looking for patterns.



Max Barret



Webecoist.com

Systems Thinking...

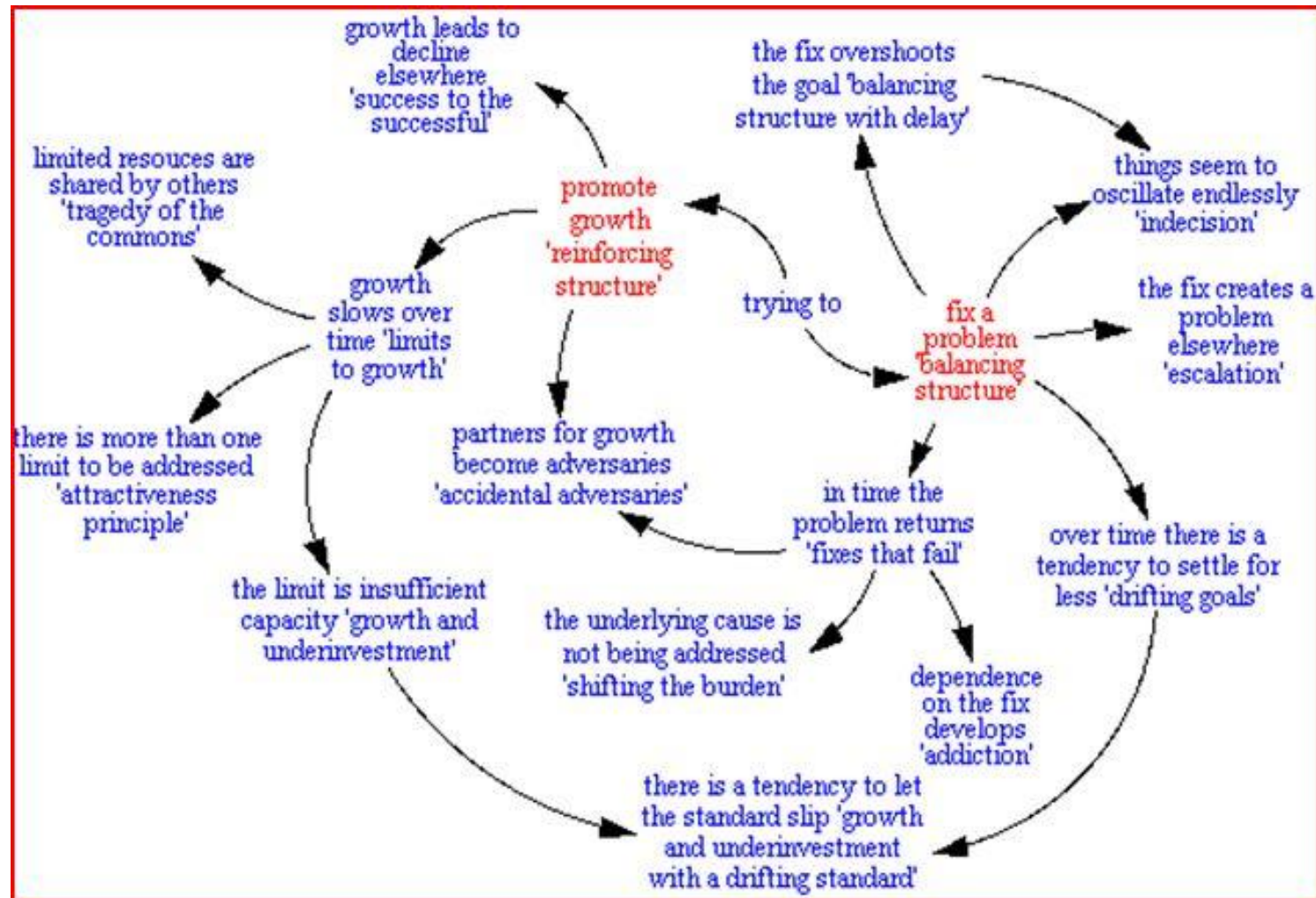
Helps us understand feedback structures that change systems over time.



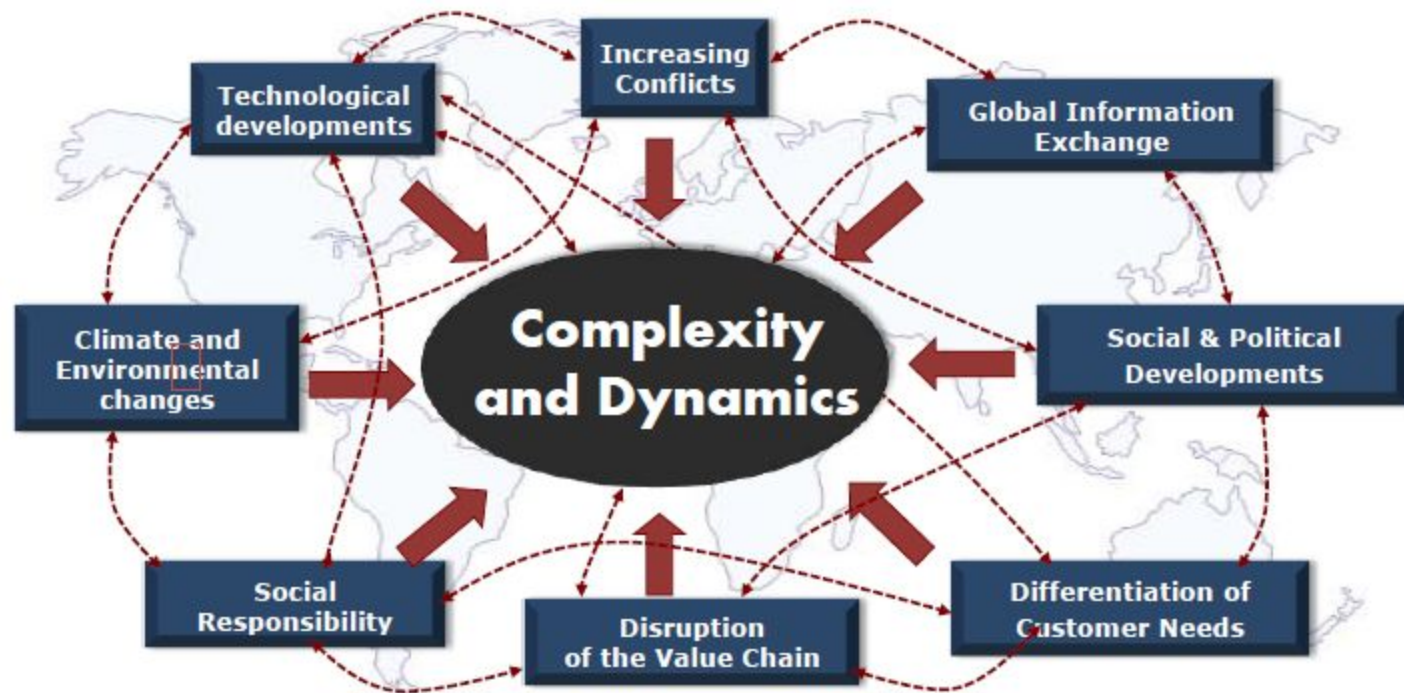
River Fractal - Héctor Garrido

Systems Thinking...

Helps us understand results of our decisions...



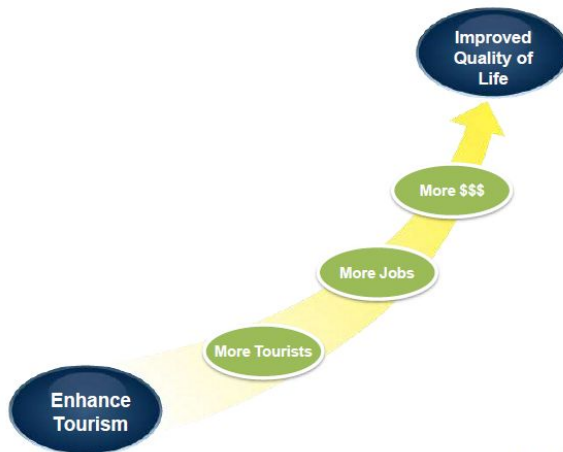
Why Systems Thinking?



Dealing with complexity and coping with increasing dynamics has become the main challenge in project and program management

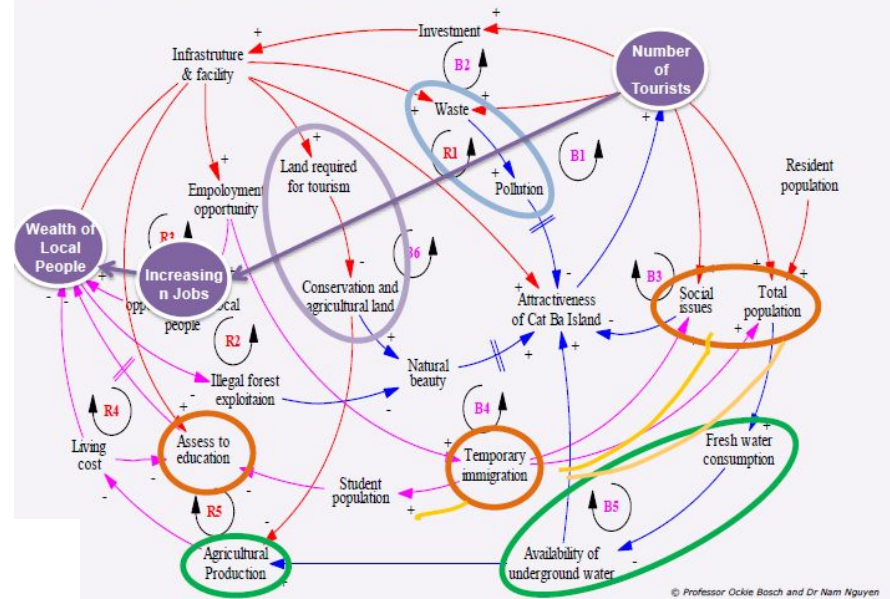
The capacity to redesign in **systems and sustainability terms**, will increasingly be what society and employers will require from new people entering the workforce

LINEAR THINKING



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SYSTEMS THINKING



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